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BRANCH: CSM

SECTION: B

COURSE NAME: FULL STACK WEB DEVELOPMENT

CODE: 23CM4121

TITLE OF THE EXPERIMENT:

To understand and implement the fundamental building blocks of HTML and HTML5 for creating static web pages.

DESCRIPTION(TASK-1) :-

This experiment brings a college's digital front door to life through the "College Information Portal" — a single HTML5 page built like a set of building blocks stacked one on top of another. Each block demonstrates a distinct concept: semantic tags (header, nav, section, article, footer) frame the page's skeleton the way corridors and departments frame a real campus; formatted text, ordered and unordered lists, and hyperlinks bring the college's story and offerings to life; a structured table and images showcase student records and campus visuals; and a fully interactive registration form — complete with radio buttons, checkboxes, dropdowns, and the newer HTML5 input types (date, number, color, range) — mirrors the kind of admission form a real student would fill out. Rounding it off, embedded audio and video elements demonstrate how modern browsers handle media natively, without needing any external plugins. Together, these pieces turn a plain page into a working, well-organised digital portal.

PROGRAM:-

```

<!DOCTYPE html>
<!-- Declares this as an HTML5 document -->
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Information Portal</title>
</head>
<body>

<!-- ===== PART A: TEXT FORMATTING, LISTS & LINKS
===== -->

<section id="about">
  <p>Welcome to our college portal. ABC Engineering College<sup>&reg;</sup>
    <!-- <sup> raises text above the line — used here for the ® symbol,
      which is always written slightly higher than normal text -->
    was established in 1998 and offers <b>quality education</b>,
    <!-- <b> makes text bold, just to highlight a keyword -->
    <i>modern infrastructure</i>, and <u>excellent placements</u>.
    <!-- <i> italicizes, <u> underlines — pure text formatting,
      no real "meaning" attached, just visual style -->
    The college is currently ranked 3<sup>rd</sup> among private engineering
    <!-- <sup> again — this time for the ordinal suffix "rd" in "3rd",
      which is always written smaller and raised in real writing -->
    colleges in the state.
  </p>

  <h2 id="courses">Courses Offered</h2>
  <!-- Ordered list: sequence/ranking matters here (as displayed on the
    admission brochure, courses are listed in order of intake priority) -->
  <ol>
    <!-- <ol> = Ordered List. Browser numbers items automatically (1,2,3...) -->
    <li>Computer Science Engineering</li>
    <li>Electronics Engineering</li>
    <li>Mechanical Engineering</li>
    <li>Chemical Engineering</li>
  </ol>

  <h2>Facilities</h2>
  <!-- Unordered list: no ranking implied, just a set of facilities -->

```

```
<ul>
  <!-- <ul> = Unordered List. Browser just adds bullet points, no numbering -->
  <li>Central Library</li>
  <li>Sports Complex</li>
  <li>Hostel</li>
  <li>Chemistry Laboratory</li>
</ul>
```

```
<p>In the Chemistry Laboratory, Chemical Engineering students carry out
experiments involving compounds such as H<sub>2</sub>SO<sub>4</sub>
<!-- <sub> lowers text below the line — needed here because chemical
      formula numbers are always written smaller and below the letter -->
(sulphuric acid titrations) and study the behaviour of gases such as
CO<sub>2</sub>, as part of their core lab curriculum.</p>
```

```
<p><small><sup>1</sup>As per the State Engineering Colleges Ranking
<!-- <small> makes footnote-style text smaller, and <sup> marks the
      "1" as a footnote reference number, same idea as in a printed book -->
Survey, 2025.</small></p>
```

```
<p>
  <a href="https://www.college-example.edu" target="_blank">Visit College Website</a> |
  <!-- <a href="..."> creates a clickable link.
      target="_blank" makes it open in a NEW browser tab
      instead of navigating away from the current page -->
  <a href="https://www.google.com" target="_blank">Google</a>
</p>
</section>
```

```
<!-- ===== PART B: TABLES AND IMAGES ===== -->
```

```
<section id="students">
  <h2>Student Details</h2>
  <table border="1" cellpadding="8">
    <!-- <table> holds tabular data. border and cellpadding are only
          for visibility while learning — real projects use CSS instead -->
    <tr>
      <!-- <tr> = Table Row -->
      <th>Roll No</th>
      <!-- <th> = Table Header cell — bold and centered by default,
```

used only in the first row to label columns -->

```
<th>Name</th>
<th>Branch</th>
<th>Email</th>
</tr>
<tr>
  <td>101</td>
  <!-- <td> = Table Data cell — holds the actual values -->
  <td>Niharika</td>
  <td>CSE</td>
  <td>niharika@example.com</td>
</tr>
<tr>
  <td>102</td>
  <td>Saranya</td>
  <td>ECE</td>
  <td>sara@example.com</td>
</tr>
<tr>
  <td>103</td>
  <td>Divya</td>
  <td>Chemical</td>
  <td>divya@example.com</td>
</tr>
</table>
```

<h2>College Logo & Campus</h2>

<!-- alt is NOT just decoration — it's what shows up if the image

fails to load, and what a screen reader speaks aloud for

visually impaired users. It should always describe the image. -->

</section>

<!-- ===== PART C: STUDENT REGISTRATION FORM ===== -->

<section id="register">

<h2>Student Registration Form</h2>

<form>

<label>Name:</label>

```
<input type="text" name="sname" required><br><br>
<!-- "required" is HTML5 built-in validation — the browser itself
      blocks submission if this field is left empty, no JS needed -->
```

```
<label>Email:</label>
<input type="email" name="semail" required><br><br>
<!-- type="email" checks that the text looks like a valid email
      (has an @ and a domain) before allowing submission -->
```

```
<label>Password:</label>
<input type="password" name="spass" required><br><br>
<!-- type="password" hides the typed characters as dots -->
```

```
<label>Gender:</label>
<input type="radio" name="gender" value="Male"> Male
<input type="radio" name="gender" value="Female"> Female<br><br>
<!-- Both radios share name="gender" — that's what links them
      together as ONE group, so only one can ever be selected -->
```

```
<label>Hobbies:</label>
<input type="checkbox" name="hobby" value="Reading"> Reading
<input type="checkbox" name="hobby" value="Sports"> Sports<br><br>
<!-- Checkboxes work independently of each other, so ticking one
      does NOT untick the other — multiple selections are allowed -->
```

```
<label>Branch:</label>
<select name="branch">
  <!-- <select> + <option> = a dropdown menu, saves screen space
        compared to listing all choices as radio buttons -->
  <option value="CSE">CSE</option>
  <option value="ECE">ECE</option>
  <option value="MECH">MECH</option>
  <option value="CHEM">Chemical</option>
</select><br><br>
```

```
<label>Comments:</label><br>
<textarea rows="4" cols="30"></textarea><br><br>
<!-- <textarea> allows multi-line free text, unlike <input type="text">
      which only allows a single line -->
```

```

    <input type="submit" value="Submit">
    <!-- Sends all form data (to a server, in a real app) -->
    <input type="reset" value="Reset">
    <!-- Clears every field back to its original blank/default state -->
</form>
</section>

<!-- ===== PART D: HTML5 FEATURES ===== -->
<section id="notice">
    <!-- <section> groups related content together as one thematic block -->
    <article>
        <!-- <article> = content that could stand alone and still make sense
            on its own, like a notice or a news post -->
        <h3>Latest Notice</h3>
        <p>Semester exams start from next month. Chemical Engineering students
            should collect their lab coats before the practical session.</p>
    </article>

    <h2>Media</h2>
    <audio controls src="welcome.mp3"></audio><br>
    <!-- controls = shows the built-in play/pause/volume bar automatically,
        no custom player needed -->
    <video controls width="300" src="campus_tour.mp4"></video>

    <h2>HTML5 Input Types</h2>
    <form>
        Email: <input type="email"><br>
        Date of Birth: <input type="date"><br>
        <!-- type="date" gives a calendar picker instead of a plain text box -->
        Age: <input type="number" min="17" max="30"><br>
        <!-- type="number" restricts input to numbers and can enforce a range -->
        Favourite Color: <input type="color"><br>
        <!-- type="color" opens a color-picker swatch -->
        Rate our campus tour: <input type="range" min="1" max="5">
        <!-- type="range" gives a slider instead of typing a number -->
    </form>
</section>
<footer>

```

```
<!-- <footer> marks the closing section of the page, e.g. copyright info -->
<p>&copy; 2026 ABC Engineering College</p>
</footer>
</body>
</html>
```

OUTPUT:-

